

# Git Hands-on Labs (HOL 1–5) Implementation Guide

## Overview

This document summarizes the Git concepts and practical implementation completed across HOL 1 to HOL 5.

## HOL 1 – Git Basics

Configured Git with user.name and user.email, initialized a repository using **git init**, created welcome.txt, checked repository status using **git status**, staged files with **git add**, committed changes with **git commit**, connected a remote repository using **git remote add origin**, and pushed code using **git push**.

**Concepts:** Working directory, staging area, local repository, remote repository, commits.

## HOL 2 – .gitignore

Created a **.gitignore** file to ignore unnecessary files and folders. Added rules such as **\*.log**, **log/**, and **\*.bak**. Verified ignored files using **git status**.

**Concepts:** Ignored files, repository cleanliness, preventing temporary files from being tracked.

## HOL 3 – Branching & Merging

Created a new branch using **git branch** and **git checkout**, added new content, committed branch-specific changes, switched back to master, compared branches using **git diff**, merged with **git merge**, viewed history using **git log --oneline --graph --decorate**, and deleted merged branches using **git branch -d**.

**Concepts:** Branch isolation, feature development, merge workflow.

## HOL 4 – Merge Conflict Resolution

Created conflicting changes to the same file (hello.xml) on two branches, observed Git conflict markers, manually resolved the conflict by editing the file, staged the resolved file with **git add**, completed the merge using **git commit**, and updated **.gitignore**.

**Concepts:** Merge conflicts, conflict markers (<<<<<<, =====, >>>>>>), manual resolution, merge commits.

## HOL 5 – Remote Repository

Verified clean working tree, added GitHub remote, pushed local commits to GitHub, verified branch tracking, and understood why nested repositories can produce 'untracked directory' messages in a parent repository.

**Concepts:** Remote repositories, origin, tracking branches, git push, git pull, nested Git repositories.

## Important Commands

```
git init
git status
git add .
git commit -m "message"
git branch
git checkout branch
git checkout -b newbranch
```

git merge branch  
git diff  
git log --oneline --graph --decorate  
git remote add origin URL  
git remote -v  
git pull origin master  
git push origin master

## Key Learning Outcomes

- Initialize and configure Git.
- Track files and create commits.
- Ignore unwanted files using .gitignore.
- Create, switch, merge, and delete branches.
- Resolve merge conflicts.
- Connect local repositories to GitHub.
- Push and pull changes.
- Understand nested Git repository issues and best practices.